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Keyhole dynamic behavior, joint properties under pulsed laser welding of Mg/Al heterogeneous ultrathin alloys

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Highlights

- Explored the dynamic behavior of keyhole under the action of pulsed laser.
- Superhydrophobic joints were prepared by pulsed laser.
- Analyzed the fracture behavior of joints after mechanical stretching.

Abstract

In this research, a pulsed laser was used for welding joining of Mg/Al heterogeneous ultrathin alloys, and the effects of different laser scanning rates on the stability of the keyhole were analyzed by capturing the keyhole dynamic behavior with an ultrahigh-speed camera. It was found that the keyhole morphology is affected by a combination of pulse spacing and scanning rate, and that the generation and dissipation of metal vapors lead to keyhole expansion and collapse, which affects the stability of the molten pool and weld uniformity. Due to the discontinuity of the pulsed laser, the stirring effect of the keyhole on the molten pool is more obvious, and the existence of the stirring effect of the keyhole makes the distribution of the heat source in the molten pool more uniform. As a result of the combined improvements, a corrosion-resistant hydrophobic joint (contact angle up to 134.4°) was prepared, and the maximum tensile strength of the joint was increased by 99 % under specific parameters.

Previous

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Keywords

Nanosecond pulsed laser welding; Mg/Al lap joints; Microinterfaces; Keyhole behavior; Joint properties; Interfacial bonding

1. Introduction

The growing academic and industrial interest in advanced alloy materials has driven significant research efforts toward developing lightweight welding solutions for component mass reduction in transportation sectors [[1], [2], [3], [4], [5]]. Structural alloys including magnesium (Mg), aluminum (Al), titanium (Ti) alloys, and stainless steel have gained prominence in engineering applications owing to their exceptional mechanical properties and industrial significance [[6], [7], [8]]. While joining dissimilar alloys presents technical complexities, such hybrid connections demonstrate substantial potential for meeting diverse design specifications and operational requirements in extreme environments.

As a critical protective component in launch vehicle architecture, the rocket fairing requires materials with exceptional thermomechanical performance to safeguard payloads against atmospheric aerodynamic forces and thermal loads during ascent

[9,10]. Current fairing designs predominantly employ aluminum alloys due to their favorable strength-to-weight ratio, corrosion resistance, and cost-effectiveness [11,12]. Nevertheless, the aerospace industry is witnessing increased adoption of advanced magnesium alloys, which offer innovative pathways for structural mass optimization while maintaining operational reliability [13]. Magnesium alloys present distinct advantages as ultra-light structural materials, possessing a density 36 % lower than aluminum and 78 % below conventional steels [14,15]. This non-ferrous metal demonstrates superior specific strength and hardness characteristics, enabling enhanced load-bearing capacity, fatigue resistance, and operational durability compared to aluminum counterparts [16,17]. Notably, magnesium alloys exhibit improved damping capacity, significantly reducing vibration-induced structural stresses [18]. Mass reduction analyses reveal that substitution of 100 kg aluminum alloy with magnesium counterparts in launch vehicle construction yields approximately 30 kg system-level weight savings. This weight differential underscores the necessity for developing reliable dissimilar joining techniques between these alloys.

The thickness of the thinnest part of the rocket fairing is about 0.3 mm, as an emerging connection technology, nanosecond pulsed laser welding has emerged as a pivotal joining technology for ultra-thin metallic components, attributed to its precision energy delivery, rapid thermal cycling, and localized heat input. These characteristics mitigate thermal distortion risks in thin-walled geometries while achieving weld depths comparable to conventional methods [19,20]. Furthermore, the technology demonstrates unique capabilities in fabricating surface-engineered microstructures. A feature particularly advantageous for aerospace systems requiring tailored surface functionalities alongside structural reliability.

Cai et al. [21] explored the keyhole dynamics behavior and molten pool flow in a laser-MIG welding mode, and they proposed a method to make the interfacial forces more balanced by suppressing the keyhole depth fluctuations. Ma et al. [22] used numerical simulations to propose that the dynamics and flow behavior of the molten pool and keyhole in laser welding can produce obvious fluctuating tendencies, which can have a significant impact on the quality of the laser welding process. Dongre et al. [23] proposed that nanosecond pulse welding technology can change the contact angle of metal surfaces, and thus prepare hydrophobic surfaces by adjusting process parameters. Wen et al. [24] explored the mechanism of the keyhole instability on the molten pool depth under different pulsed laser parameters, and the results showed that the pulse frequency has a direct effect on the keyhole depth fluctuation. Wu et al. [25] successfully joined 0.1 mm steel and 0.6 mm magnesium alloys using nanosecond pulsed laser technology to explore the effect of scanning speed on their joints. Feng et al. [26] used nanosecond pulsed laser technology to achieve the joining of 0.4 mm 316 stainless steel and 0.2 mm 6063 aluminum alloy. And this study observed the presence of elemental diffusion at the welded joints and the generation of intermetallic compounds (IMC) layers at the joint interface, and the generation of IMCs seriously affected the joint strength. Experiments by Wang et al. [27] explored the effect of nanosecond pulsed laser process on the generation of microtextures on the surface of aluminum alloys, emphasizing that the process parameters have an influence on the grain structure and texture formation on the alloy surface. Dong et al. [28] used nanosecond pulsed laser process to join ultrathin metal foils with a thickness of only 50 μm, and experimentally obtained welds with very few defects and high-strength joints. The experimental results show that controlling the pulse laser energy density by changing nanosecond pulsed laser welding parameters can greatly improve the microstructure characteristics and mechanical properties of the joints. Feng et al. [29] investigated the effect of different laser pulse frequencies on nanosecond pulsed laser welding of aluminum alloy and stainless steel of 0.2 mm. The experiment successfully joined the two metals and achieved good joint quality. It was also shown that the proper laser pulse frequency can reduce the energy effect of a single pulse, resulting in less porosity and cracking defects in the joint. Xie et al. [30] used laser oscillation mode for hybrid laser-arc welding of Al–Ti dissimilar butt joints, and they investigated the generation mechanism of multilayer IMCs at the joint bonding interface and suggested that the local turbulence inside the molten pool is conducive to the optimization of the composition of the IMCs layer at the interface and the reduction of its thickness.

In summary, in recent years, there have been many research reports on keyhole behavior and interfacial bonding, but most of the studies have been carried out in the continuous welding mode of medium-high thickness metal plates and more focused on the determination of mechanical properties of the joints. Therefore, in this research, a discontinuous type pulsed laser was used to study the welding of ultrathin metal plates, and the dynamic behavior of the keyhole during the welding of Mg/Al dissimilar ultrathin alloys in this mode was deeply investigated. In addition, hydrophilic joints and hydrophobic joints were prepared and the mechanical tensile properties of the joints were tested in this research, which provides new thinking for pulse laser welding of other dissimilar ultra-thin alloy plates.

2. Experimental process

2.1. Materials preparation

This experiment uses AZ31B magnesium alloy and 6061 aluminum alloy with a thickness of 0.2 mm, before the experiment, the two alloys are cut into a rectangular sheet of 50mm × 10 mm and placed in a 75 % alcohol solution and ultrasonically cleaned for 5min to wash off the alloy surface stains. The chemical compositions of the two alloys are shown in Table 1, Table 2.

Table 1. The standard chemical composition of AZ31B.

Mg	Al	Zn	Mn	Si	Fe	Cu	Ni
Bal.	3.1	0.9	0.32	0.012	0.0021	0.009	0.0009

Table 2. The standard chemical composition of 6061.

Al	Mg	Mn	Si	Fe	Cu	Cr	Zn	Ti
Bal.	0.6–1.2	0.15	0.40	0.70	0.15	0.05–0.35	0.25	0.1

To achieve effective removal of surface oxides and impurities prior to welding, systematic pretreatment procedures must be implemented on specimens. Initial mechanical treatment involves abrasion with coarse-grit sandpaper to eliminate inherent oxide layers and machining marks. Subsequent multi-stage refinement is conducted using sequentially graded sandpapers from 800 to 1500 mesh. Under the operation of a rotary polishing apparatus at 300 rpm, a diamond abrasive suspension (2.5 µm particle size) is applied to enhance the frictional efficiency of the polishing cloth. Continuous polishing cycles are maintained until a mirror-finish surface is attained. Post-polishing chemical treatment entails immersion in Kroll's reagent ($HNO_3 : HF : H_2O = 1 : 2 : 50$)for approximately 60 s.

2.2. Nanosecond pulse welding program

Following preliminary experimental investigations, the Mg–Al lap joint configuration was optimized to mitigate vaporization risks associated with magnesium alloy's low boiling point. As illustrated in Fig. 1 (a), the magnesium substrate was positioned beneath the aluminum component in the overlap arrangement, with a 10 mm × 10 mm overlap area established at the interface. Considering the power constraints inherent to nanosecond pulsed laser systems, a concentric multi-ring welding strategy was implemented to overcome insufficient heat accumulation. The welding pattern comprised progressively diminishing concentric circles, initiating from an outer diameter of 3.6 mm and terminating at an inner diameter of 0.2 mm (Fig. 1 (b)). This outward-to-inward progression facilitates controlled thermal superposition through successive laser passes, thereby achieving cumulative energy deposition while preventing excessive instantaneous power delivery. The laser energy density was quantified using simulation, as shown in Fig. 1 (c). In the welding area, the highest energy density reached 95J/ mm^2.

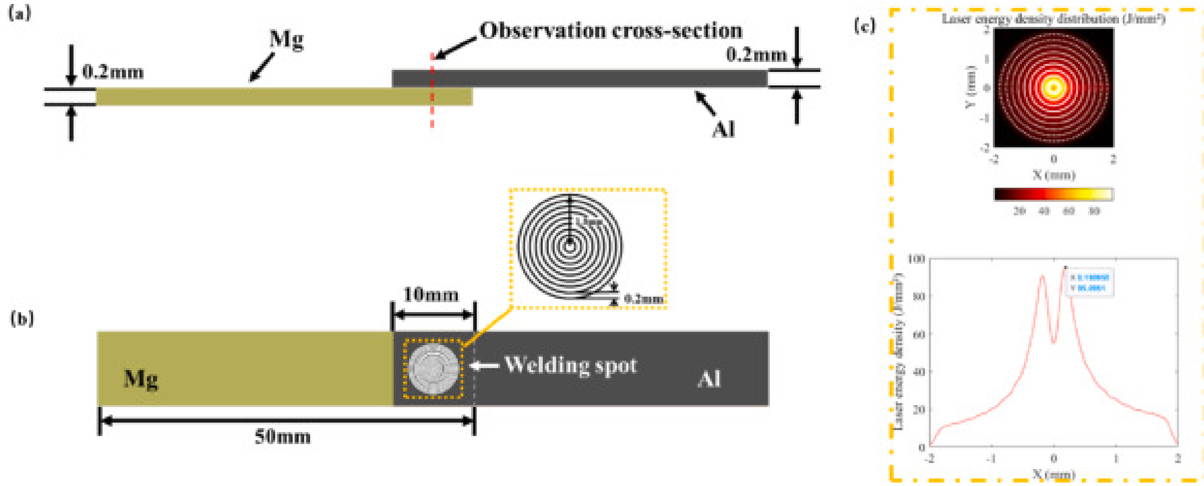

Fig. 1. Schematic diagram of nanosecond pulsed laser welded magnesium/aluminum lap joint: (a) sample lap configuration, (b) weld pattern,(c) laser energy density distribution.

Fig. 2 shows an self-designed ultra-thin metal sheet clamping welding platform, which uses four bolted connections to clamp the platen in order to hold the sample through the platen. So that the fitting between the two samples is sufficiently adequate. The laser is used to weld the surface of the sample through the reserved welding area. An ultra-high-speed camera is used to record the changes in the sample surface during the welding process.

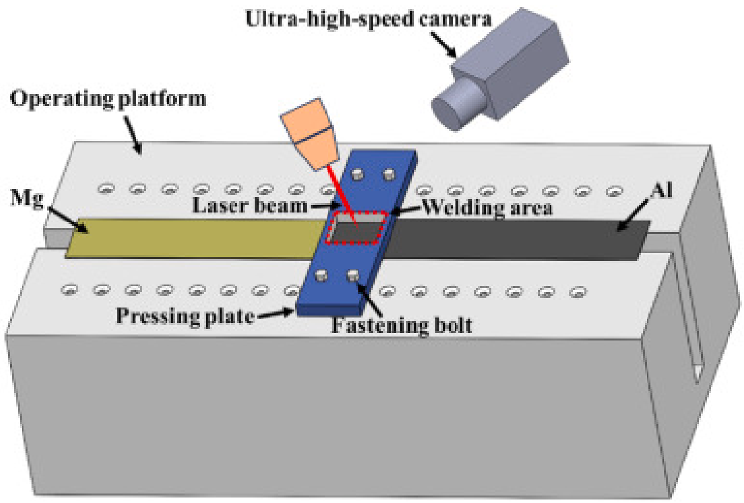

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Fig. 2. Self-designed ultra-thin metal sheet clamping welding platform.

This experiment uses a nanosecond pulsed laser produced by China Jingwei Laser Company. The equipment model is JW-F30W, its wavelength is 1064 mm, the beam diameter is 0.1 mm. Through the computer to draw the scanning path of the laser, the computer to process the data transmitted to the laser transmitter, then by the laser transmitter to launch a parallel laser beam. The beam is reflected through the X-axis mirrors and Y-axis mirrors to special positions on the galvanometer. The parallel beams are focused by the mirrors and then transmitted to the processing platform to complete a specific laser scanning path. The schematic diagram of the nanosecond laser light path is shown in Fig. 3. The nanosecond pulsed laser welding parameters are shown in Table 3.

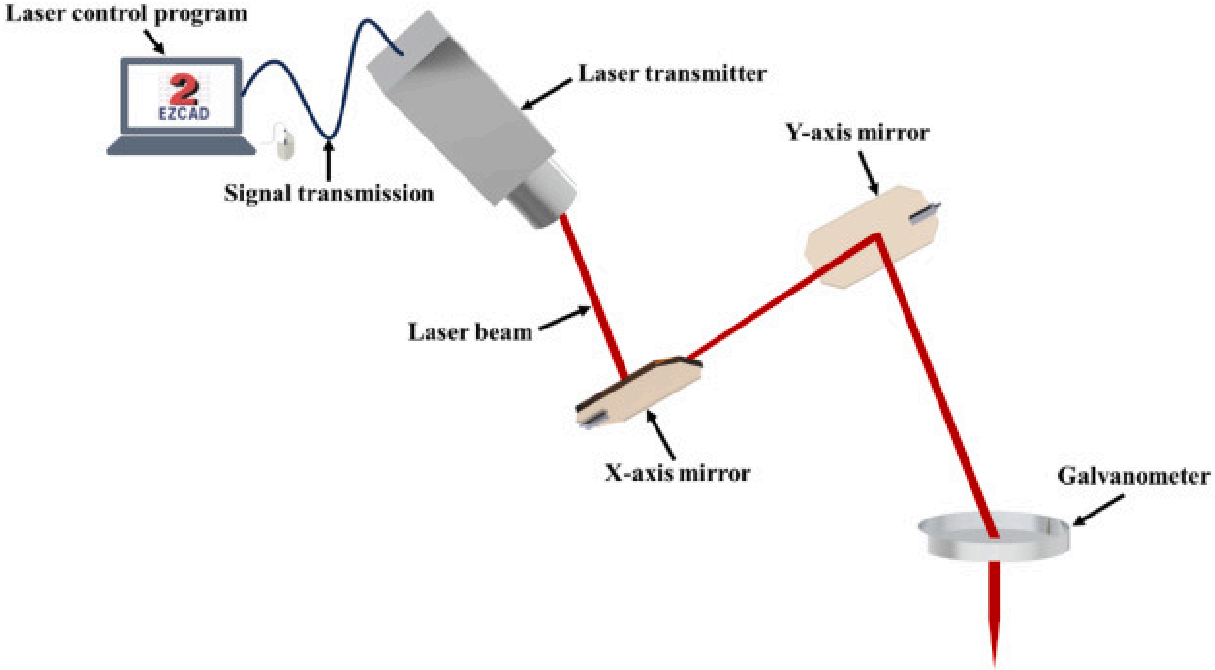

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Fig. 3. Schematic diagram of nanosecond laser light path.

Table 3. Nanosecond pulsed laser welding parameters.

Welding parameters	**Value**
Power(W)	25
Frequency(kHz)	20
Sweep line spacing(mm)	0.2
The pulse width(ns)	10
Welding speed(mm/s)	10,20,30,40,50,60,70

Welding parameters	Value
Defocus amount	0

2.3. Mechanical property test methods

Post-welding evaluation combined multi-scale characterization techniques. Macroscopic weld morphology was analyzed using optical microscopy, while microstructural features of surface and cross-sectional areas were investigated through scanning electron microscopy coupled with elemental mapping spectroscopy. High-speed imaging captured dynamic melt pool behavior during the welding process. Surface wettability was quantified through static contact angle measurements employing the sessile drop technique. Phase composition analysis was performed using X-ray diffraction. Mechanical properties were assessed via standardized tensile shear testing. Specimen alignment was ensured through 0.2 mm thickness precision spacers. Statistical reliability was achieved by reporting mean maximum load values from triplicate tests, with outliers excluded through consistency validation. Fig. 4 illustrates the Schematic diagram of tensile shear test.

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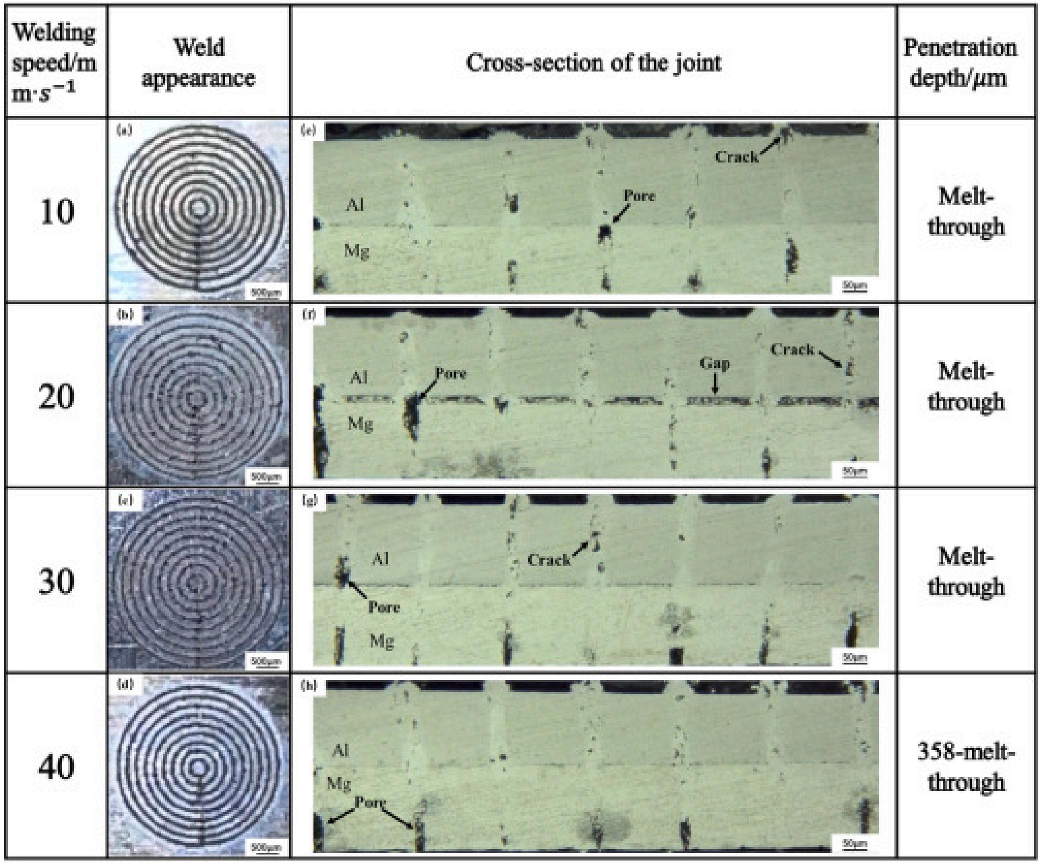
Fig. 4. Schematic diagram of tensile shear test.

3. Experimental results

3.1. Analysis of Mg/Al pulsed laser weld formation

The weld surfaces at different laser scanning speeds are shown in Fig. 5(a–d) and Fig. 6(a–c). The weld surface shows a clear laser scanning trajectory, but exhibits severe spattering at laser scanning speeds of 20 mm/s-30 mm/s, which is consistent with the excessive "peaks" observed under the ultra-depth-of-field microscope at this laser scanning speed interval. Fig. 5(e–h) and Fig. 6(d–f) show the joint cross sections at different laser scanning speeds. As the laser scanning speed increases, the laser energy density increases, resulting in different joint penetration depths for different laser scanning speeds. The joint cross-section is mainly characterized by porosity defects, and the porosity is mainly distributed in the magnesium side, which is related to the lower boiling point of magnesium alloy [31]. Under the instantaneous high temperature, the magnesium alloy reaches a certain temperature and then vaporizes and evaporates. But due to the high solidification rate of the molten pool, a part of the magnesium vapor fails to escape in time to form the air holes. With the acceleration of the laser scanning speed, the size of the pores becomes larger, which will be further analyzed in the following text. At the laser scanning speed of 60 mm/s-70 mm/s, the two alloy joint surfaces show large gaps, which is related to the lack of heat accumulation at high laser scanning speeds. These defects such as pores and gaps will reduce the effective contact area of the joint, which will affect the mechanical properties of the joint.



Welding speed/mm·s^{-1}	Weld appearance	Cross-section of the joint	Penetration depth/μm
10	(a) 500μm	(e) Crack, Pore, Al, Mg, 50μm	Melt-through
20	(b) 500μm	(f) Crack, Gap, Pore, Al, Mg, 50μm	Melt-through
30	(c) 500μm	(g) Crack, Al, Pore, Mg, 50μm	Melt-through
40	(d) 500μm	(h) Al, Mg, Pore, 50μm	358-melt-through

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Fig. 5. Morphology of the joint at different laser scanning speeds: (a–d) weld surface, (e–h) cross section.

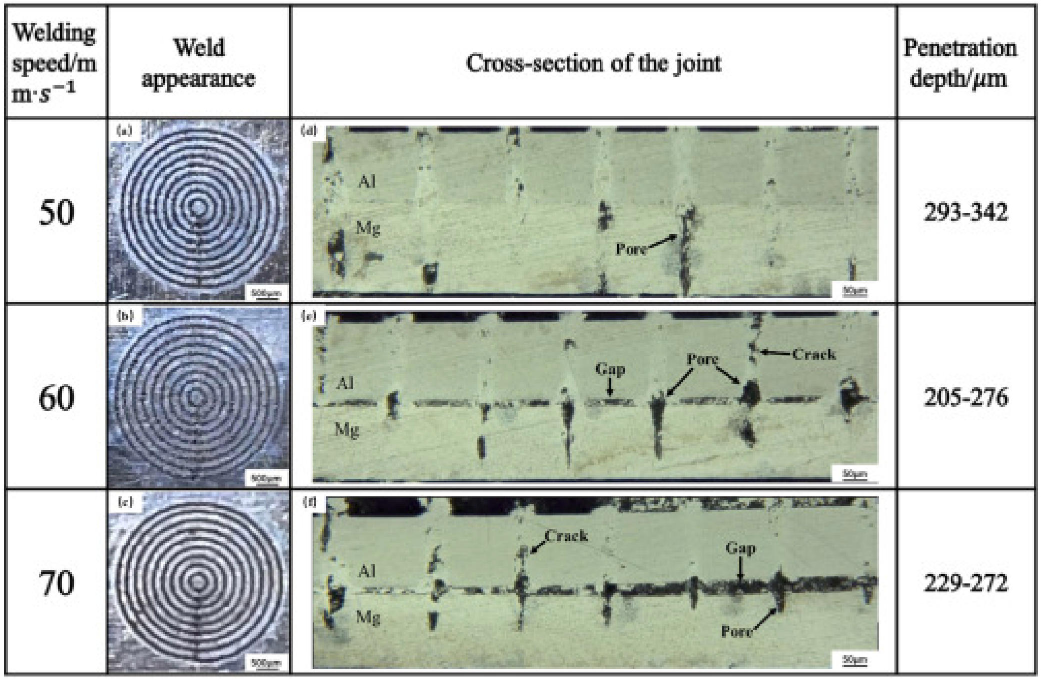

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Fig. 6. Morphology of the joint at different laser scanning speeds: (a–c) weld surface, (d–f) cross section.

To further investigate the effects of laser scanning speed on pore size and distribution, selected regions were chosen to statistically analyze the pore distribution, as illustrated in the 3D map (a) and X–Y planar map (b) of Fig. 7. In the 3D representation, the sphere sizes directly reflect pore dimensions. The 3D map reveals that larger pores predominantly cluster at higher scanning speeds. At lower scanning speeds, both the top and bottom regions primarily contain fine pores. This occurs because lower scanning speeds allow extended laser-substrate interaction per unit distance, enabling more magnesium alloy vaporization and rapid formation of a larger molten pool. The vaporized magnesium subsequently rises and releases rapidly within the alloy, while the expanded molten pool provides sufficient time for magnesium vapor escape [32], leaving only minor pores trapped internally. Conversely, at excessively high scanning speeds where full penetration is not achieved, shortened solidification time traps incompletely escaped magnesium vapor within the alloy, forming elongated macro-pores [33]. These larger pores concentrate in the bottom region and migrate upward with increasing scanning speed.

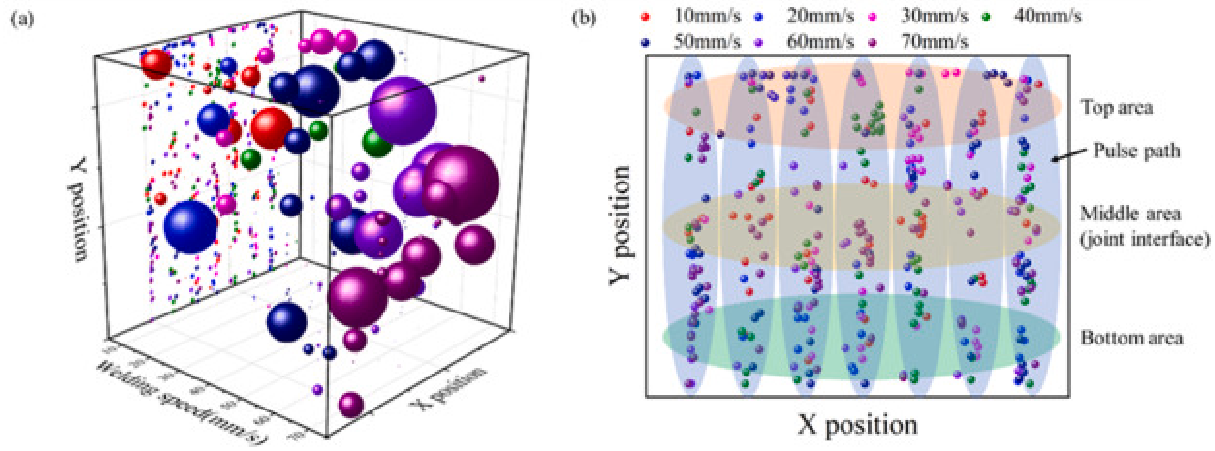

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Fig. 7. Distribution law of pores: (a) 3D view, (b) X–Y view.

The 2D mapping clearly demonstrates radial pore distribution along seven pulse trajectories. The central region (joint interface) exhibits predominant 70mm/s pores, indicating concentrated pore aggregation at the bonding interface between parent materials under this scanning speed. This phenomenon not only reduces effective joint contact area but also intensifies interface brittleness through pore clustering, ultimately compromising mechanical performance. Additionally, pores with different scanning speeds are evenly distributed in the top and bottom regions.

3.2. Dynamic behavior of keyhole under pulsed laser action

When a high-energy laser acts on the material surface, it causes rapid temperature rise, leading to melting or even vaporization, thereby forming distinct high-temperature and high-luminance regions. The intense thermal radiation emitted by these regions can be clearly captured by an ultra-high-speed camera. By continuously monitoring the dynamic changes of these high-luminance regions, the dynamic behavior of the keyhole can be inferred to evaluate the stability of the welding process [34]. Specifically, the expansion of the keyhole is manifested as the rapid enlargement and extension of the high-luminance region, while the collapse of the keyhole is often accompanied by the violent contraction and darkening of the high-luminance region.

Since pulsed lasers are characterized by multiple non-continuous pulses acting on the surface of the material to form a molten pool to complete the weld, the morphology of the keyhole is influenced by the pulse spacing in conjunction with the laser scanning rate. The generation of metal vapor disperses the molten metal in the focal area of the laser, resulting in the formation of a keyhole. As the laser continues to act, more metal vapor is generated, causing more molten metal to be dispersed, resulting in the expansion of the keyhole. Conversely, due to the discontinuous characteristics of the nanosecond pulsed laser, when the pulsed laser ceases to act, the production of metal vapor decreases and is insufficient to continue dispersing the molten metal, allowing the molten metal to backfill the keyhole region, thereby triggering the collapse of the keyhole. The shorter keyhole expansion-collapse cycle means the larger the keyhole fluctuation per unit time, the more unstable the interaction between the molten metal region (molten pool) and the keyhole, which further affects the uniformity of the weld structure and weakens the joint strength. Therefore, the keyhole stability at different laser scanning rates can be inferred from the frequency and magnitude of keyhole changes, thus providing a strategy to reduce thermal defects in the joint.

Two different laser scanning rates were selected to investigate the keyhole dynamic behavior under pulsed laser action, with the start and end sections of the entire welding process chosen as research objects. The ultra-high-speed image of the beginning section of the welding under 10mm/s is shown in Fig. 8 (a1-a6), and the ultra-high-speed image of the end section of the welding under 10mm/s is shown in Fig. 8 (b1-b6). The keyhole expansion-collapse cycle in the welding start section is shorter than that in the end section; in other words, the keyhole behavior in the welding end section is more stable. Additionally, the welding process is accompanied by metal plasma spatter, and this phenomenon occurs at a lower frequency in the end section.

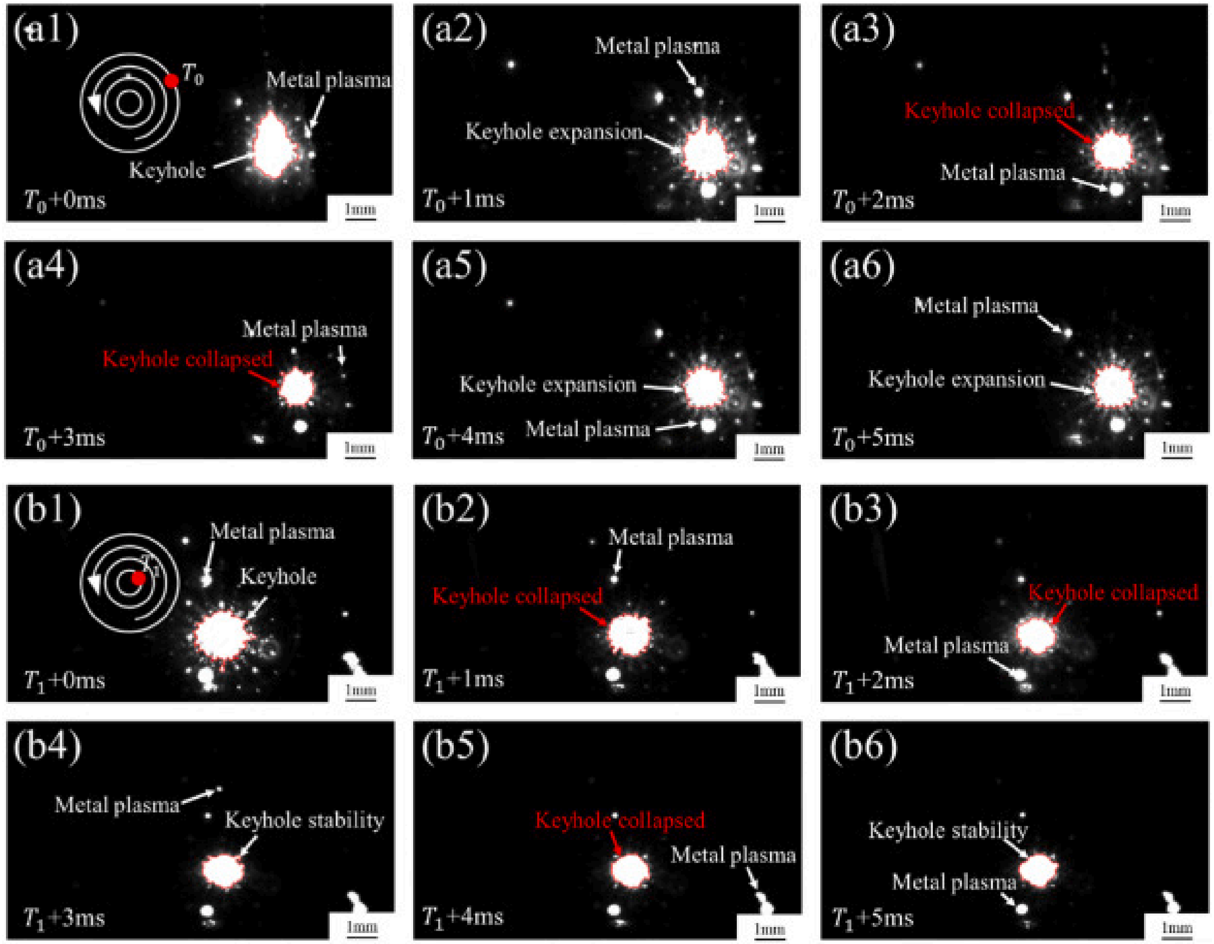

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Fig. 8. Ultra-high-speed images at 10mm/s: (a1-a6) beginning segment keyhole images, (b1-b6) end segment keyhole images.

The ultra-high-speed images of the welding start section at 50mm/s are shown in Fig. 9 (a1-a6), and those of the welding end section at 50mm/s are shown in Fig. 9 (b1-b6). The keyhole expansion-collapse cycle in the start section at 50mm/s is shorter than that in the end section, with the keyhole behavior in the end section being more stable. Compared with the 10mm/s rate, the size of the high-luminance region in both the start and end sections at 50mm/s increases significantly, and keyhole expansion/collapse occurs more frequently, while spatter phenomena also increase markedly. Therefore, it can be inferred that frequent keyhole expansion and collapse at high scanning speeds severely disturb the molten pool, leading to more and more dispersed metal plasma spatter.

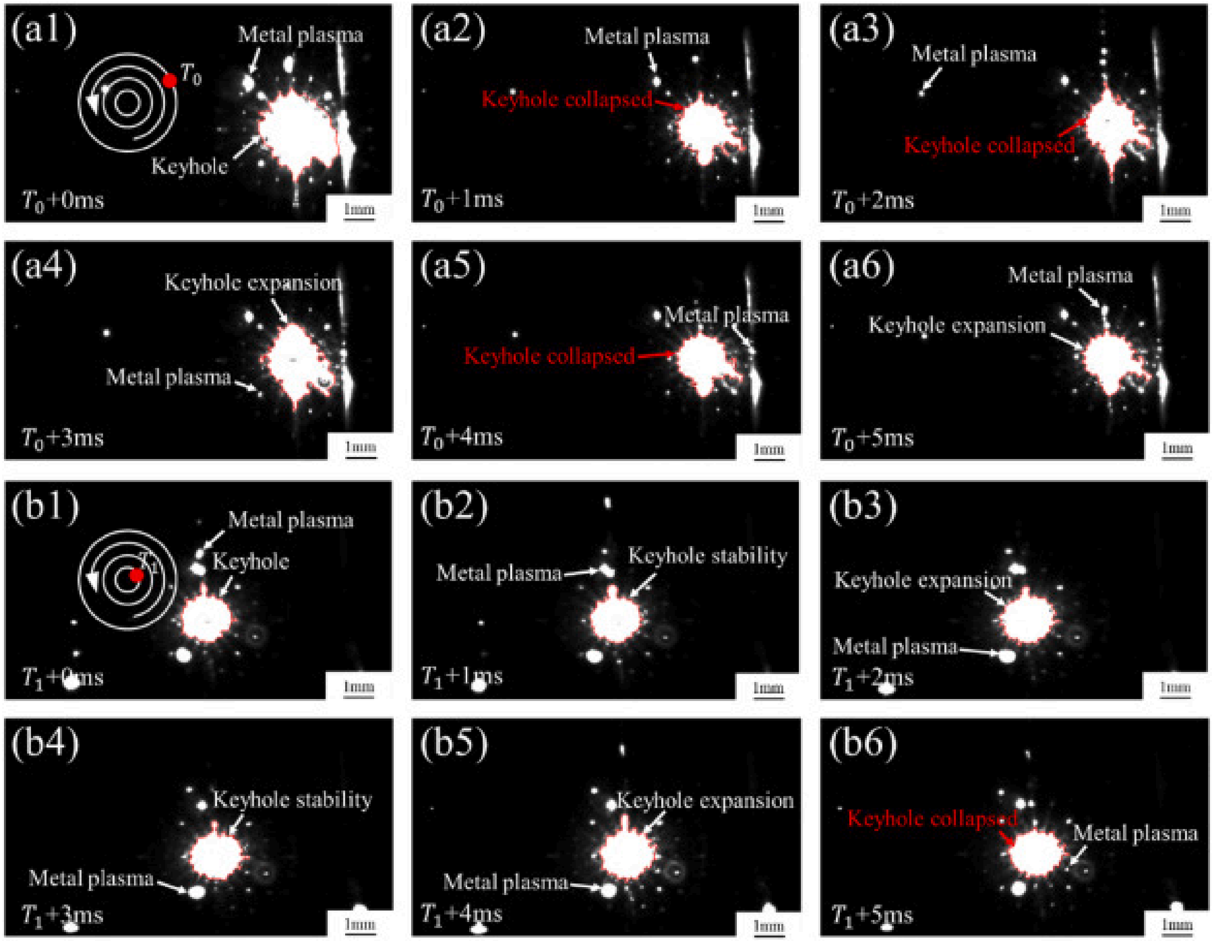

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Fig. 9. Ultra-high-speed images at 50 mm/s: (a1-a6) beginning segment keyhole images, (b1-b6) end segment keyhole images.

Table 4 below summarizes the keyhole dynamic behavior data, defining the dynamic cycle as the state change between expansion→collapse and collapse→expansion. The overall dynamic stability of the keyhole is reflected by quantitatively statistics the length and number of dynamic cycles. At 10 mm/s, 2 dynamic cycles were captured in the welding start section, while no dynamic cycles were detected in the end section; at 50 mm/s, 3 dynamic cycles were captured in the start section and 1 dynamic cycle in the end section, with the dynamic cycles being generally shorter under 50 mm/s conditions, indicating poorer keyhole stability at high scanning speeds. Overall, under both scanning speed conditions, keyhole stability in the welding end section is better than that in the start section.

Table 4. Table of dynamic changes in keyhole at different times.

Case	Times	Keyhole status	Expansion-Collapsed cycle time
10 mm/s	T_0 +0 ms~T_0 +1 ms	Expansion	/
	T_0 +1 ms~T_0 +2 ms	Expansion→Collapsed	2 ms
	T_0 +2 ms~T_0 +3 ms	Collapsed→Collapsed	/
	T_0 +3 ms~T_0 +4 ms	Collapsed→Expansion	2 ms
	T_0 +4 ms~T_0 +5 ms	Expansion→Expansion	/
	T_1 +0 ms~T_1 +1 ms	Collapsed	/
	T_1 +1 ms~T_1 +2 ms	Collapsed→Collapsed	/
	T_1 +2 ms~T_1 +3 ms	Collapsed→Stability	/
	T_1 +3 ms~T_1 +4 ms	Stability→Collapsed	/
	T_1 +4 ms~T_1 +5 ms	Collapsed→Stability	/
50 mm/s	T_0 +0 ms~T_0 +1 ms	Collapsed	/
	T_0 +1 ms~T_0 +2 ms	Collapsed→Collapsed	/
	T_0 +2 ms~T_0 +3 ms	Collapsed→Expansion	3 ms
	T_0 +3 ms~T_0 +4 ms	Expansion→Collapsed	1 ms
	T_0 +4 ms~T_0 +5 ms	Collapsed→Expansion	1 ms

Case	Times	Keyhole status	Expansion-Collapsed cycle time
	T_1 +0 ms~T_1 +1 ms	Stability	/
	T_1 +1 ms~T_1 +2 ms	Stability→Expansion	/
	T_1 +2 ms~T_1 +3 ms	Expansion→Stability	/
	T_1 +3 ms~T_1 +4 ms	Stability→Expansion	/
	T_1 +4 ms~T_1 +5 ms	Expansion→Collapsed	5 ms

3.3. Microstructure analysis of Mg/Al joint interface

Fig. 10 shows the SEM images of the welds at different laser scanning speeds. In this study, two groups of low and high laser scanning speeds were selected for further analysis. In both Fig. 10(a) and (b), mechanical self-locking structures of varying heights was observed on the base material joint surface, and this structure was more obvious at the high laser scanning speed. This structure was also observed in the experiments of Zhu et al. [35]. In Fig. 10(c) and (d), different color gradients are observed, and the closer to the center of the weld the darker the color becomes. This different color gradient divides the weld into three zones, which are the reactive layer, the heat-affected layer, and the base material layer. In the reaction layer in Fig. 10(d), bright banded and striped structures were observed, requiring further elemental analysis.

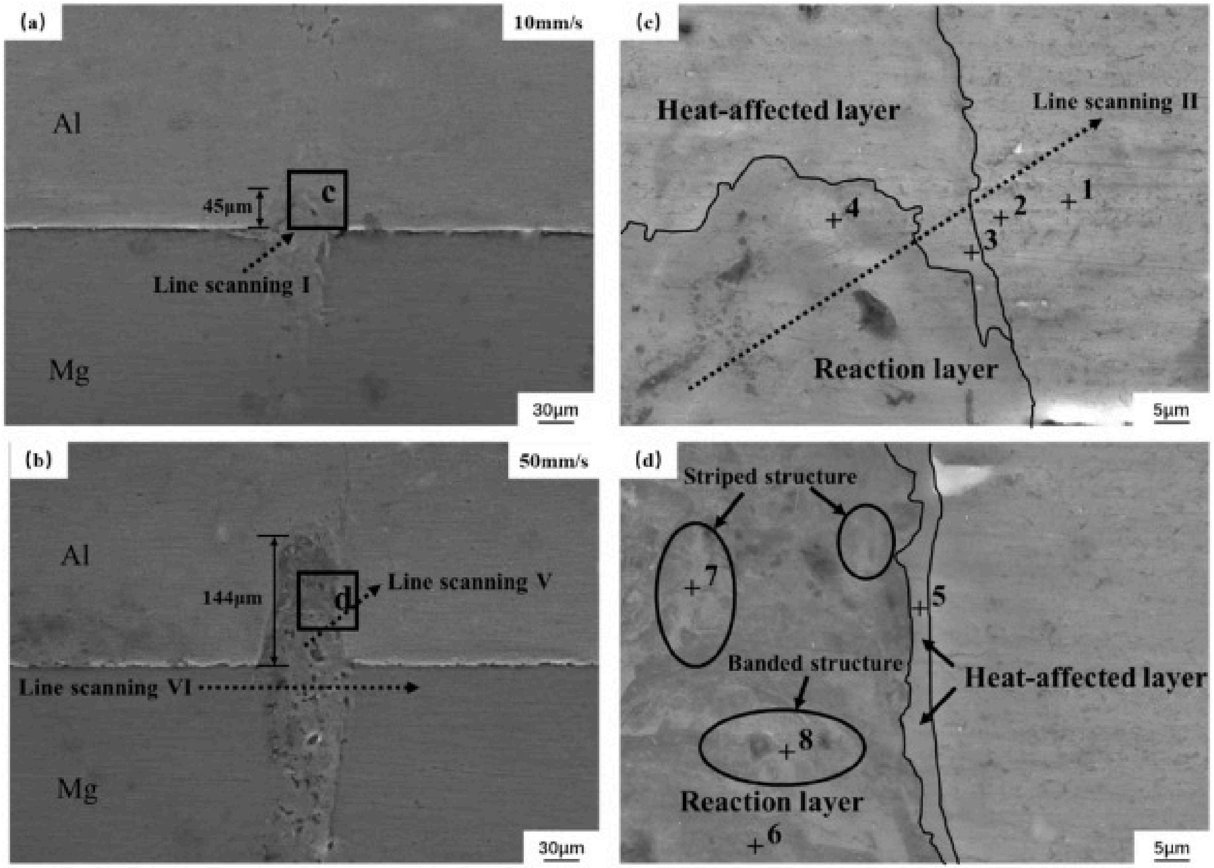

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Fig. 10. SEM images of welds with different laser scanning speeds: (a) 10mm/s, (b) 50mm/s, (c–d) magnified images of the marked area.

Table 5 shows the results of EDS point scans at different points in Fig. 10. According to the results, all the points at the bright structure were detected to contain IMCs, and the closer to the center of the weld, the higher the ratio of Mg elements in IMCs. The white striped structure point 3 and point 7 detected Mg and Al atomic ratio is about 1:1, it can be inferred that the striped structure is MgAl. white band structure point 8 detected Mg and Al atomic ratio is about 17:12, it can be inferred that the band structure is $Mg_{17}Al_{12}$. It is worth mentioning that the Mg element content is very high at point 6, which indicates that the magnesium-rich domain has been formed in the center of the aforementioned reverse “V” structure, and the element diffusion phenomenon is obvious, which is conducive to the full bonding of the two base materials. The content of Al element in point 5 is about 25.94 %, and it can be inferred from the phase diagram of Mg–Al binary alloy [36] that the main composition of this point is a mixed phase of $Mg_{17}Al_{12}$ and α-Mg.

Table 5. Scanning results of different points on the weld seam.

Point	**Mg(At%)**	**Al(At%)**	**Possible phases**
1	4.7	95.3	Al
2	12.09	87.91	αAl
3	17.84	82.16	Mg_2Al_3 +αAl
4	52.28	47.72	MgAl
5	74.06	25.94	$Mg_{17}Al_{12}$ +αMg
6	89.3	10.7	αMg
7	55.5	44.5	MgAl
8	58.68	41.32	$Mg_{17}Al_{12}$

To systematically investigate the phase composition within the welded joints, cross-sectional specimens were subjected to X-ray diffraction analysis following standard metallographic preparation. The resulting XRD pattern, presented in Fig. 11 reveals distinct intermetallic compounds including Mg_2Al_3 and $Mg_{17}Al_{12}$ phases which corroborate the predictions outlined in Table 5. Notably, the analytical limitations of current spatially resolved XRD techniques necessitate whole-joint examination rather than localized micro-interface between Mg/Al constituents. This constraint accounts for the observed single-peak intensity anomalies arising from matrix material interference. Furthermore, the XRD analysis detected non-stoichiometric crystalline phases ($Al_{0.58}Mg_{0.42}$ and $Al_{0.56}Mg_{0.44}$) that were not identified through energy-dispersive spectroscopy. The presence of these metastable phases suggests non-equilibrium solidification characteristic of the laser welding process.

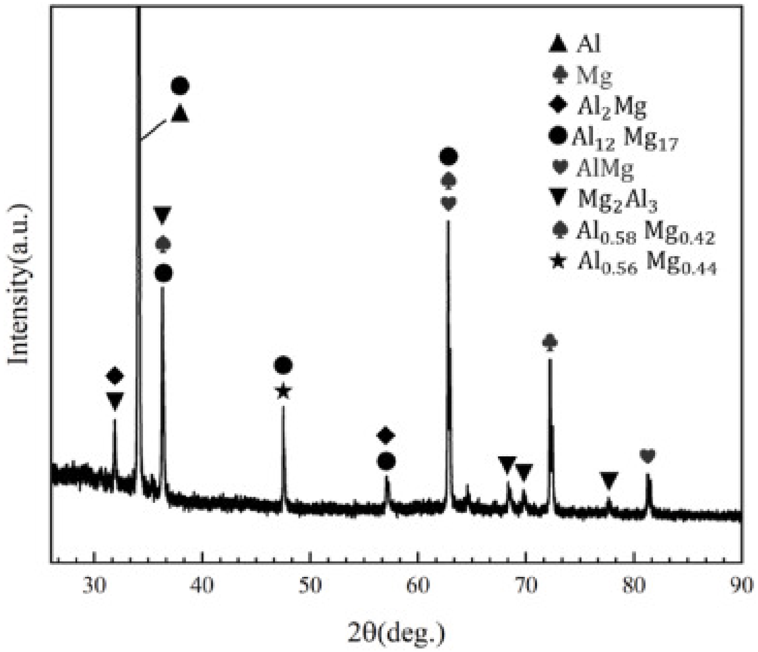

Fig. 11. Joints phase composition diagram.

Fig. 12 shows the EDS mapping results of the welds at different laser scanning speeds. The results show that at different laser scanning speeds, the penetration of Mg elements into the aluminum side was observed, which further explains the reason for the formation of the mechanical self-locking structures in Fig. 10. In Fig. 12(b) and (c), it is observed that the elemental penetration appears as a multilayered band structure, which was also observed in the study of Feng et al. [29]. The molten metal at the center of the weld is extruded by the recoil pressure of magnesium vapor [37], and this process causes the Mg element to float upward while moving radially along the nanosecond pulsed laser beam. The molten metal is affected by the Marangoni effect and is stirred several times inside the molten pool as the laser processing continues, and generating the multilayered band structure in the region close to the edge of the weld.

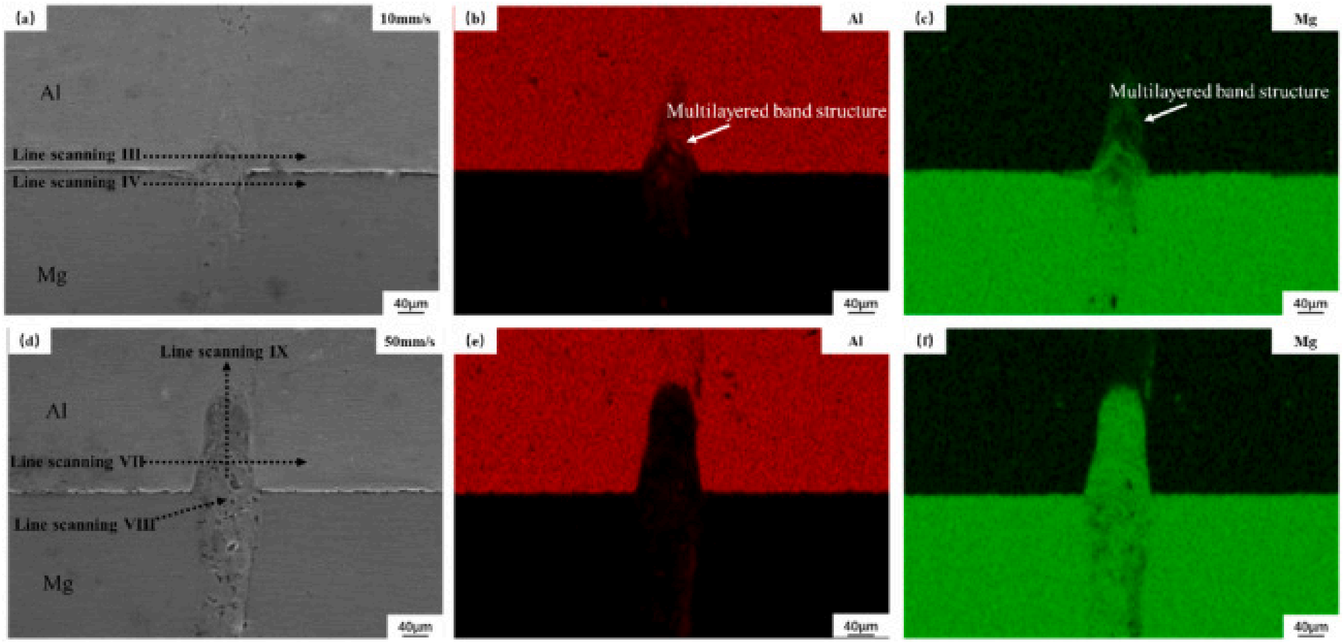

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Fig. 12. EDS mapping results for welds at different laser scanning speeds: (a–c) 10mm/s, (d–f) 50mm/s.

To further analyze the variation law of elemental content in different zones, Fig. 13 shows the line scanning results of the marked straight lines in Fig. 10, Fig. 12. According to the different elemental content, the weld can be divided into the base metal zone, heat-affected zone, and weld zone. The elemental content of the base metal zone changes smoothly and does not involve elemental diffusion phenomenon. The elemental content in the heat-affected zone varies greatly, with obvious elemental diffusion, and the stable Mg–Al elemental transition zone is formed there, which further indicates that the heat-affected zone is more prone to produce IMCs. The thickness of the IMCs layer is directly proportional to the width of the heat-affected zone. The elemental content variation in the weld zone is relatively smooth, and the elemental distribution is diametrically opposite to that of the base metal zone, as shown in Fig. 13(b), (c), (g), (i). This indicates that the elements have been sufficiently diffused in this region. Taking the line scanning results of the approximate area under different laser scanning speeds as shown in Fig. 13(c)–(g), the width of the heat-affected zone with the laser scanning speed of 10mm/s is wider than that of 50mm/s. This is because at low laser scanning speeds, the heat input per unit time is higher, and the heat accumulation in the center weld is large. Leading to obvious thermal diffusion phenomenon, enhanced metallurgical reaction at the interface, and larger heat-affected zone. Therefore, it can be inferred that when the laser scanning speed is 10 mm/s, the thickness of IMCs layer at the weld is larger than that at the weld when the laser scanning speed is 50 mm/s. The larger the thickness of IMCs layer is, the more brittle the joint is, and the worse the mechanical properties of the joint are. Fig. 13(a)–(d), (f), (h) shows the results of EDS line scanning in the magnesium side. The variation of Mg and Al elements is not obvious, and there is no phenomenon that the content of elements on the opposite side is larger than that on the same side. This indicates that the diffusion of elements in the magnesium side is not obvious. This is due to the relatively high density of the aluminum alloy in the molten state, which is not easy to sink and diffuse due to the impact of the pulsed laser beam. Besides, the aluminum alloy is prone to coarsening of the parent material grain under high heat input, which affects the elemental flow [38].

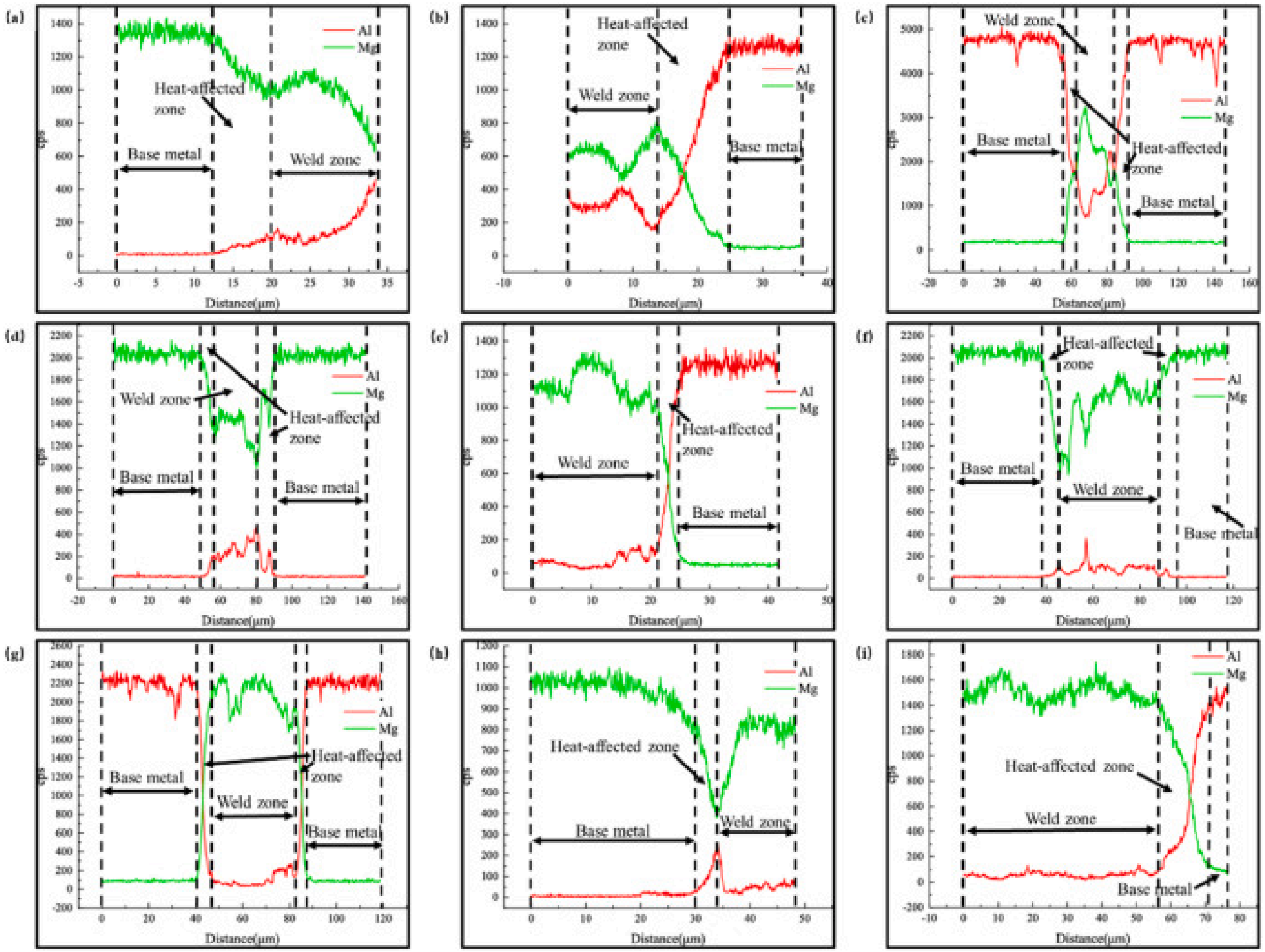

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Fig. 13. Line scanning results of the marked straight lines in Fig. 10, Fig. 12: (a) line scanning I, (b) line scanning II, (c) line scanning III, (d) line scanning IV, (e) line scanning V, (f) line scanning VI, (g) line scanning VII, (h) line scanning VIII, and (i) line scanning IX.

3.4. Properties analysis of Mg/Al joints

The current research shows that the presence of IMC layer weakens the corrosion resistance of joints, and, previous experiments have demonstrated that the interfacial microstructure has a significant effect on the corrosion resistance of joints [39,40]. In this experiment, through the test joints of hydrophobicity (repulsion) and hydrophilicity, to judge the corrosion resistance of the joints: hydrophobicity reflects the joints of water repulsion phenomenon is strong, unit time, the joints by water spread wetting the smaller the area, the joints surface of the corrosive media such as water shielding effect of the joints is bigger, so that the joints have good corrosion resistance. The quantification of hydrophobicity and hydrophilicity for joints is expressed in the form of contact angle: contact angle greater than 90° presents hydrophobicity and contact angle less than 90° presents hydrophilicity. In this paper, the droplet method was used to determine the contact angle changes at different laser scanning rates for the interfaces microstructure of filled circular, as shown in Fig. 14, Fig. 15.The experimental images of the water droplets not contacting the interface, the instant of water droplet contacting the interface, the water droplets after 60s of wetting and spreading at the interface, and the water droplets after 120s of wetting and spreading at the interface were recorded sequentially in the figure. At 30 mm/s, the contact angle of the water droplets on the aluminum alloy surface showed 134.4°, and remained at 100.5° in 120s with the wetting and spreading time, showing excellent hydrophobicity. All other scanning rates show hydrophilicity.

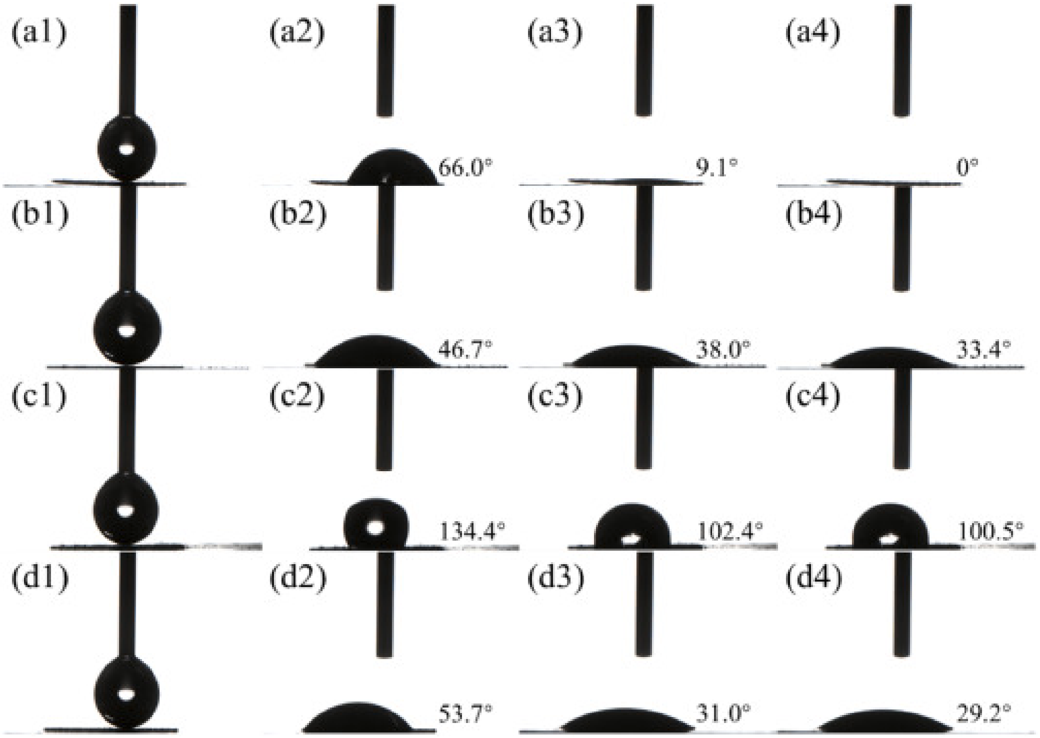

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Fig. 14. Water droplet wetting and spreading behavior: (a–d) are the contact angle changes under scanning rate of 10 mm/s-40mm/s in order, (1–4) are the state diagrams of the water droplet when it has not contacted the surface, and when it has contacted the surface for 0s, 60s, and 120s in order.

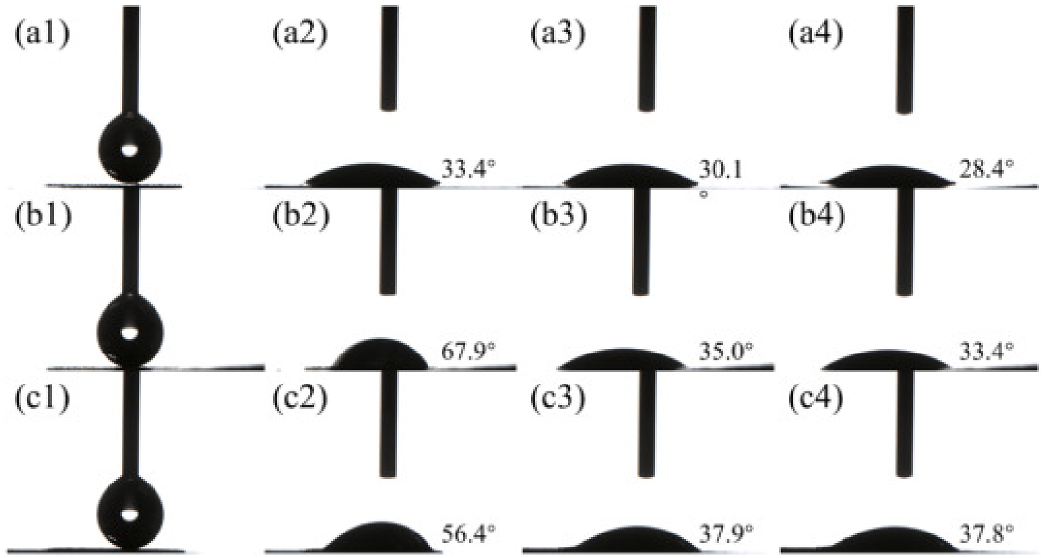

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Fig. 15. Water droplet wetting and spreading behavior: (a–c) are the contact angle changes under scanning rate of 50 mm/s-70mm/s in order, (1–4) are the state diagrams of the water droplet when it has not contacted the surface, and when it has contacted the surface for 0s, 60s, and 120s in order.

The mechanical tensile properties directly reflect the ability of the joint to withstand the maximum load. In this research, the maximum tensile force of the joints was tested at different laser scanning speeds, and the results are shown in Fig. 16(a). The maximum value was obtained at the laser scanning speed of 50mm/s, with a maximum value of 101.5N. The minimum value was obtained at the laser scanning speed of 10mm/s, with a minimum value of 51N. The maximum tensile force of the joints was increased by about 99 % compared with the minimum value. At laser scanning speeds from 10mm/s to 50mm/s, the maximum tension tends to increase. At laser scanning speeds from 50mm/s to 70mm/s, the maximum pulling force tends to decrease. With the further reduction of the laser scanning speed, the heat input is insufficient for the two parent materials to be fully joined, so that the mechanical properties of the joint are at a low level. After several experimental analyses, the fracture mode of the joint was magnesium-side button pullout failure (BPF) at laser scanning speeds of 10 mm/s-40mm/s. This fracture occurred at the outermost weld seam on the magnesium side, and there was no fracture at the joint interface. At the laser scanning speed of 50 mm/s-70mm/s, the joint fracture mode is interface failure (IF). This fracture occurred at the joint interface, and the outermost welds on the magnesium and aluminum sides did not fracture. As shown in Fig. 16(b), each layer of circular rings on the surface of the joint has a certain height difference from the surface, which can increase the amount of air trapped between the droplet and the surface, promoting the formation of Cassie Baxter state [41]. At a laser scanning speed of 30mm/s, the maximum contact angle is 134.4°, indicating hydrophobicity.

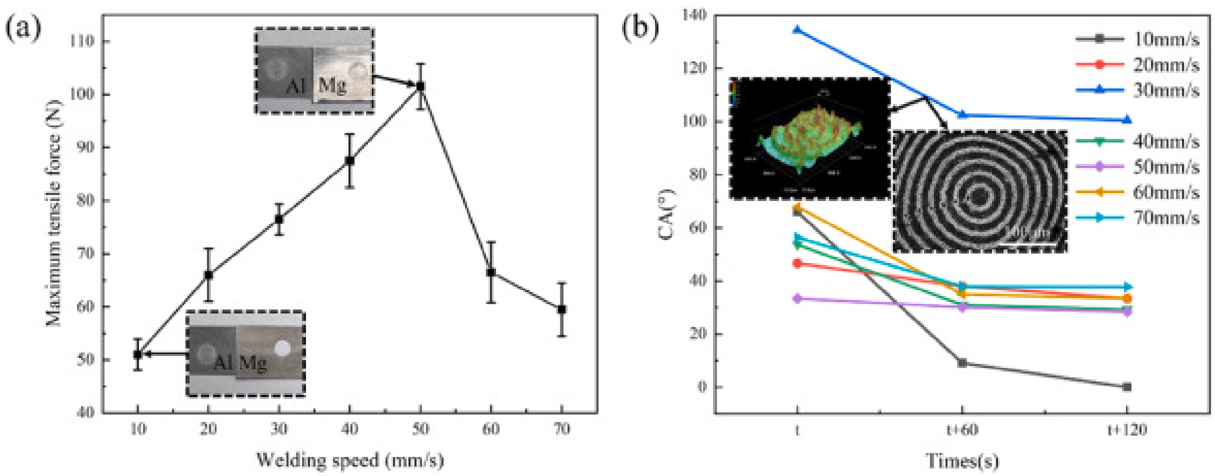

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Fig. 16. Joint properties: (a) mechanical properties, (b) wetting spreading properties.

The two sets of cases show different fracture patterns. In Fig. 17(a–g), there is a significant warpage angle at low laser scanning speeds, and this warpage behavior extends from the outermost weld towards the center and the two parent materials are in a connected state from the beginning to the end, which further indicates that the weakest region of the joint in this case lies in the outermost weld portion. Since the base material did not fracture, the connection properties of the base material were not fully realized, and the tensile strength of the joint would be limited in this case. The reason for this phenomenon was that the laser was scanning from the outside to the inside, and the heat build-up in the outermost weld was insufficient to weld the outermost weld adequately. Furthermore, the differential thermal gradients between internal and external regions induce transient thermal stresses during rapid heating/cooling cycles, promoting crack initiation at stress concentration zones [31]. Subsequent tensile loading propagates these initial defects along weld boundaries through a progressive crack extension mechanism [25], ultimately resulting in interfacial failure. At high laser scanning speeds, the two parent materials were pulled off and the scanning path on the magnesium side was clearly visible, without producing obvious traces of warpage deformation. As shown in Fig. 17(b–d), a long structure in the torn state is observed, and the structure is accompanied by bright shapes, and as analyzed previously, there is a diffusion of Mg elements to the Al side, so it is presumed that the main components of this structure are brittle IMCs and MgO. As shown in Fig. 17(f–h), a large number of holes are observed on the surface of the magnesium side. Comparison with Fig. 17(a–d) shows that bright aggregates appeared in the weld on the aluminum side surface, so it is inferred that the missing molten metal and accumulated oxides in the holes were attached to the weld surface on the aluminum side together after the fracture, which suggests that the fracture behaviors of the two parent materials in this case occurred firstly in the IMCs and oxides-rich region inside the magnesium matrix.

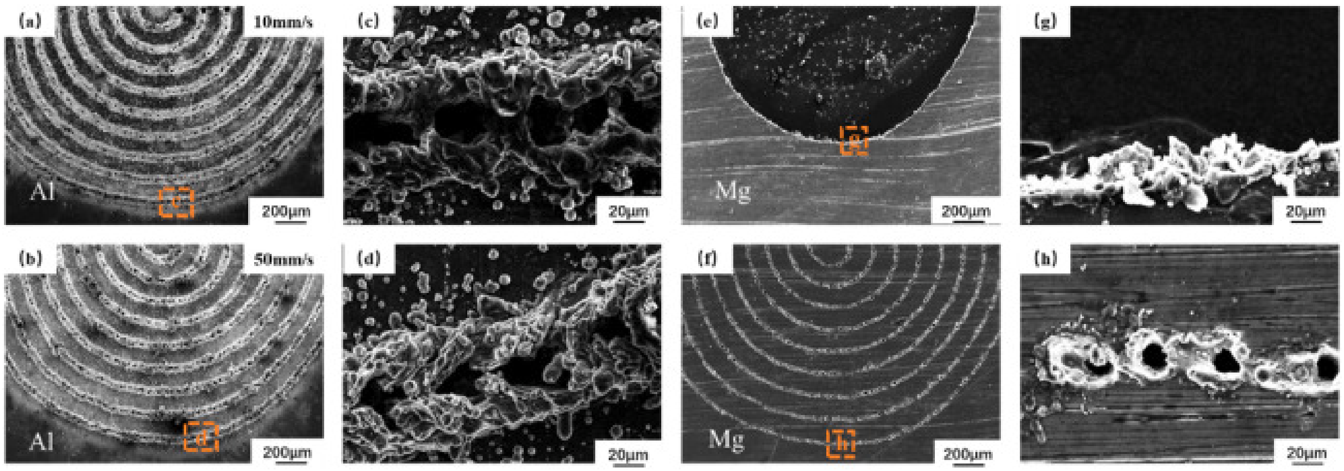

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Fig. 17. Fracture microscopic morphology of the joints at laser scanning speeds of 10mm/s and 50mm/s: (a–b) fracture surfaces on the aluminum side, (e–f) fracture surfaces on the magnesium side, (c–d) (g–h) localized magnified views.

4. Discussion

4.1. Keyhole stirring effect under pulsed laser action

Combined with the experimental data of the previous ultra-high-speed image, it is found that there is a certain dynamic behavior of the keyhole under the action of pulsed laser, as shown in Fig. 18. Under the action of a single pulse, the high energy density characteristic of the pulsed laser single point causes the internal part of the material to vaporize rapidly, forming the phenomenon of keyhole expansion, as shown in Fig. 18(a and b). When the pulsed laser is closed, the heat-affected zone shrinks in a very short interval of time, and since the internal temperature does not satisfy the conditions for keyhole formation, the surrounding paste-like molten metal begins to fill upward, resulting in the keyhole collapse phenomenon, as shown in Fig. 18(c). When the pulsed laser impacts the material again, in addition to expanding the keyhole, under the combined action of multiple pulsed lasers, a stirring phenomenon of molten metal-metal vapor occurs inside the material, as shown in Fig. 18 (d、d1): The keyhole and the surrounding molten metal exhibit mutual motion between them, with the keyhole being subjected to a resistance force $\mathbf{F_r}$ in the opposite direction of its motion [42], The surrounding molten metal is subjected to a reverse force $\mathbf{F_d}$ from the keyhole. the larger this interaction force, the more pronounced the stirring phenomenon [30].

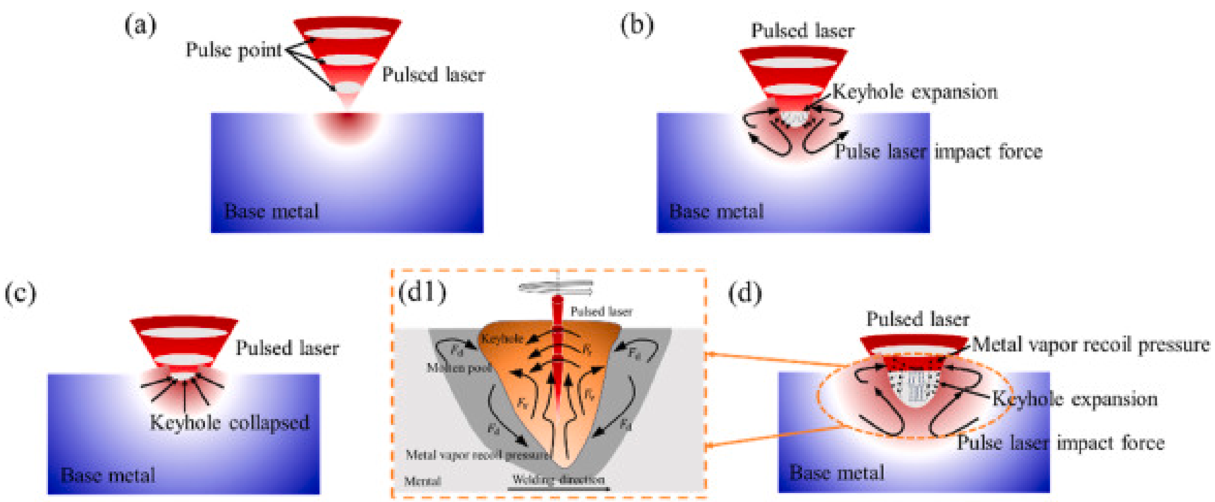

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Fig. 18. Schematic diagram of the dynamic behavior of the keyhole under pulsed laser action: (a) initial stage, (b) pulsed laser acting on the material surface, (c) keyhole collapse, (d) keyhole expansion, (d1) keyhole-molten pool forces.

The molten pool fluid flow type and stirring intensity can be expressed by the Reynolds number Re [43]:

$$\mathrm{Re} = \frac{4\rho \mathrm{N}{\mathrm{R}_0}^2}{\mu} \tag{1}$$

$$\mu = \mu_0 \ \exp\left(\frac{\mathrm{E}_0}{\mathrm{RT}}\right) \tag{2}$$

$$\mathrm{Re} = \frac{4\rho \mathrm{N}{\mathrm{R}_0}^2}{\mu_0 \ \exp\left(\frac{\mathrm{E}_0}{\mathrm{RT}}\right)} \tag{3}$$

μ represents the dynamic viscosity, N represents the stirring frequency, which is modulated by combined pulse frequency and scanning rate, R_0 represents the keyhole radius, E_0 represents activation energy, T represents the temperature, R represents the universal gas constant, and μ_0 represents viscosity coefficient. Fluid dynamics analysis reveals that when the Reynolds number (Re) remains below 1000, the molten pool maintains predominantly laminar flow characteristics. As Re exceeds this critical threshold, a laminar-turbulent transition occurs, significantly altering momentum transfer mechanisms. Enhanced scanning rates induce proportional elevation of stirring frequency (N), thereby increasing Re values and intensifying fluid agitation. This augmented stirring effect promotes more efficient mass and energy exchange between the keyhole and molten pool, leading to improved homogenization of thermal gradients within the weld zone.

5. Conclusion

(1) The defects of weld shaping are mainly dominated by porosity. Before the weld is melted through, the porosity is mainly tiny and uniformly distributed in the joint cross-section; after the weld is melted through, the pores become larger and concentrated in the bottom area of the molten pool. The reduction of the heat input results in the pores are distributed at the joint surface of the two alloys.

(2) By comparing the keyhole expansion-collapse cycle to evaluate keyhole stability, an increase in scanning rate exacerbates the instability of the keyhole in the welding start section, while keyhole stability in the welding end section is better than that in the start section. The keyhole exhibits a stirring effect on the molten pool, which can optimize the heat source distribution within the molten pool.

(3) Unbalanced crystallization and brittle IMC component were detected at the joints, and modulation of the scanning rate can optimize the extent of the heat-affected zone and shear thinning the IMC layer to some extent.

(4) The welding pattern of the filled circle can prepare hydrophobic and hydrophilic joints with practical production significance, and the mechanical properties of the joints can be improved by 99 % by optimizing the pulsed laser.

CRediT authorship contribution statement

Xianlong Wu: Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kaihan Zhou:** Supervision, Software, Project administration. **Qingshuai Gao:** Validation, Supervision, Software. **Youbin Wang:** Visualization, Validation. **Guangyu Sun:** Methodology, Investigation. **Qingning Xue:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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